

QUESTION PAPER CODE : PSTR**Name of the Candidate:****Reg. No.****APRIL 2025 ARREAR EXAMINATIONS****B.E. ELECTRICAL AND ELECTRONICS ENGINEERING****18EE410/21EER10 PROBABILITY AND RANDOM PROCESSES****Duration : 3Hours****Maximum: 100 Marks****SPECIAL INSTRUCTIONS: Non Programmable calculator shall be permitted.****Answer ALL Questions****PART – A****(5 x 6 = 30)****CO Marks
CO1 (6)**

- A1.** A store carries flash drives with either 1 GB, 2 GB, 4 GB, 8 GB, or 16 GB of memory. The accompanying table gives the distribution of of memory in a purchased drive:

Y	1	2	4	8	16
P(y)	0.05	0.10	0.35	0.40	0.10

Identify the cumulative distribution function and the graph of cumulative distribution function.

- A2.** Let the time between two successive arrivals at the drive-up window of a local bank. If X has an exponential distribution with $\lambda=1$, identify the following: **CO2 (6)**

- a.** The expected time between two successive arrivals.
b. The standard deviation of the time between successive arrivals.

- A3.** On the basis of data reported in the article "Serum Ferritin in an Elderly Population" (*J. of Gerontology*, 1979: 521–524), the authors concluded that the ferritin distribution in the elderly had a smaller variance than in the younger adults. (Serum ferritin is used in diagnosing iron deficiency.) For a sample of 28 elderly men, the sample standard deviation of serum ferritin (mg/L) was $s_1 = 52.6$; for 26 young men, the sample standard deviation was $s_2=84.2$. Identify whether the data support the conclusion as applied to men. **CO3 (6)**

- A4.** Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal, are same against that they are not, at 5% level. **CO3 (6)**

- A5.** Show that the random process $X(t) = A \cos(at+b)$ is widely stationary if A and a are constants and b is uniformly distributed random variable $(0, 2\pi)$. **CO6 (6)**

PART – B**(70 Marks)****CO Marks
CO1 (9)**

- B1.** A library has an upper limit of 6 on the number of videos that can be checked out to an individual at one time. Consider only those who check out videos, and let X denote the number of videos checked out to a randomly selected individual. The probability mass function of X is as follows:

x	1	2	3	4	5	6
P(x)	0.30	0.25	0.15	0.05	0.10	0.15

Compute the mean, variance and standard deviation of the given distribution. Also compute $\text{Var}(2x^2+3x+9)$.

(OR)

- B2.** The article "Modeling Sediment and Water Column Interactions for Hydrophobic Pollutants" (*Water Research*, 1984: 1169–1174) suggests the uniform distribution on the interval $(7.5, 20)$ as a model for depth (cm) of the bioturbation layer in sediment in a certain region. **CO1 (9)**
- a. Calculate the mean and variance of depth.

Contd...2...

- b. Compute the cumulative distribution function of depth.
- c. Compute the probability that observed depth is at most 10. Between 10 and 15.

- B3. a.** A particular telephone number is used to receive both voice calls and fax messages. Suppose that 25% of the incoming calls involve fax messages, and consider a sample of 25 incoming calls. Compute the probability that:
- a.** At most 6 of the calls involve a fax message
 - b.** Exactly 6 of the calls involve a fax message
 - c.** At least 6 of the calls involve a fax message
 - d.** More than 6 of the calls involve a fax message
- b.** If a publisher of nontechnical books takes great pains to ensure that its books are free of typographical errors, so that the probability of any given page containing at least one such error is .005 and errors are independent from page to page, compute the probability that one of its 400-page novels will contain exactly one page with errors and at most three pages with errors.
- (OR)**
- B4. a.** Suppose the diameter at breast height (in.) of trees of a certain type is normally distributed with mean 8.8 and standard deviation 2.8 as suggested in the article "Simulating a Harvester-Forwarder Softwood Thinning"
- a.** Compute the probability that the diameter of a randomly selected tree will be at least 10 in. Will exceed 10 in.
 - b.** Compute the probability that the diameter of a randomly selected tree will exceed 20 in.?
 - c.** Calculate the probability that the diameter of a randomly selected tree will be between 5 and 10 in.?
- b.** The article "Probabilistic Fatigue Evaluation of Riveted Railway Bridges" (*J. of Bridge Engr.*, 2008: 237–244) suggested the exponential distribution with mean value 6 MPa as a model for the distribution of stress range in certain bridge connections. Let's assume that this is in fact the true model. Compute the probability that the stress range is almost 10MPa, the stress range is between 5MPa to 10MPa.
- B5. a.** A manufacturer of sprinkler systems used for fire protection in office buildings claims that the true average system-activation temperature is 130°. A sample of 9 systems, when tested, yields a sample average activation temperature of 131.08°F. If the distribution of activation times is normal with standard deviation 1.5°F, illustrate does the data contradict the manufacturer's claim at significance level 0.01.
- b.** The target thickness for silicon wafers used in a certain type of integrated circuit is 245 mm. A sample of 50 wafers is obtained and the thickness of each one is determined, resulting in a sample mean thickness of 246.18 mm and a sample standard deviation of 3.60 mm. Illustrate if the data suggest that true average wafer thickness is something other than the target value.
- (OR)**
- B6. a.** Natural cork in wine bottles is subject to deterioration, and as a result wine in such bottles may experience contamination. The article "Effects of Bottle Closure Type on Consumer Perceptions of Wine Quality" (*Amer. J. of Enology and Viticulture*, 2007: 182–191) reported that, in a tasting of commercial chardonnays, 16 of 91 bottles were considered spoiled to some extent by cork-associated characteristics. Illustrate does this data provide strong evidence for concluding that more than 15% of all such bottles are contaminated in this way. Carry out a test of hypotheses using a significance level of 0.10.

- b.** A letter in the *Journal of the American Medical Association* (May 19, 1978) reported that of 215 male physicians who were Harvard graduates and died between November 1974 and October 1977, the 125 in full-time practice lived an average of 48.9 years beyond graduation, whereas the 90 with academic affiliations lived an average of 43.2 years beyond graduation. Illustrate does the data suggest that the mean lifetime after graduation for doctors in full-time practice exceeds the mean lifetime for those who have an academic affiliation? **CO3 (8)**

- B7.** Using the multiplicative congruential method, compute the period of the generator for $a = 13$, $m = 2^6$ and $X_0 = 1, 2, 3$, and 4. **CO4 (10)**

(OR)

- B8.** Buses arrive at the bus stop at Peachtree and North Avenue according to a Poisson process with a mean of one bus per 15 minutes. Generate a random variate, N , which represents the number of arriving buses during a one-hour time slot. Now, N is Poisson distributed with a mean of four buses per hour. **CO4 (10)**

- B9.** Demonstrate a straight line fit to the following data: **CO5 (10)**

x	6	7	7	8	8	8	9	9	10
y	5	5	4	5	4	3	4	3	3

(OR)

- B10.** Calculate the mean radiation dose at an altitude of 3000 feet by fitting an exponential curve to the given data: **CO5 (10)**

Altitude(x)	50	450	780	1200	4400	4800	5300
Dose of radiation (y)	28	30	32	36	51	58	69

- B11.** If $X(t) = \mu + N(t)$, where $N(t)$ is white noise with $C(t_1, t_2) = \sigma(t_1)\delta(t_1, t_2)$, where $\sigma(t)$ is a bounded function of t and δ is the unit impulse function, prove that $\{X(t)\}$ is a mean ergodic process. **CO6 (14)**

(OR)

- B12.** The power spectrum of a WSS process $\{X(t)\}$ is given by $S(\omega) = \frac{1}{(1+\omega^2)^2}$. **CO6 (14)**
Compute its autocorrelation function $R(\tau)$ and average power.

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